ActInf Textbook Group ~ Cohort 4 ~ Meeting 11 (Chapter 5 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | *Cohort 4 ~ Meeting 11 (Chapter 5 part 1)* | *59:18*

all right welcome thank you everyone
it's August 15 23 and word cohort 4 in
our first discussion of chapter five
so
soon we will screen share
but first
any remarks that anybody wants to make
on
chapter five
just anything that stayed with them
after reading it or
feelings that they had or anything else
about chapter five
I will say uh I found this to convert
and conversely to chapter four being
rather mathematically against this one
is a little bit more of
neurobiologically dense at least for
those who maybe like are new to
um thinking about the relationship
between this and areas of the brain but
I did appreciate in this chapter there
was much more on Precision which we
didn't get too much before
um even though it was brought up many a
Time preceding chapters and I really
appreciated the relationship between
uh that was made between different forms
of precision
and different neurotransmitters
neuromodulation we found that to be very
interesting and helpful and maybe kind
of reinforcing the idea of like they're
not just probability distributions here
but also

uh Precision of each or like confidence
in each probability distributions
so my big takeaway from this chapter
cool thank you
anyone else is there any thoughts on
five
any other thoughts or any any question
that somebody wants to
lead with or we can just explore the
chapter and look at some written
questions
um I did I did want to ask uh what
everyone makes of uh
the relationship
made here I mean in preceding chapters
we're talking a lot about
um
you know kind of the differences between
viewing variables as categorical versus
continuous and it's a big kind of
bifurcation throughout many of these
chapters like we have continuous models
versus
um you know Palm DPS and
um just different ways of looking at
things but it at one point in this
chapter they point to the idea that
um there's actually a big relationship a
connection between the two that
um
and maybe what's going on in the upper
levels
of say the the brain cortical regions
and so on is that they're working with
categorical variables but in a way it's
not simply categorical variables but
rather these are like um looking at
discrete

trajectories that the continuous
variables could take
that was a nice way of maybe putting
making some kind of thread between the
two ways of looking variables that is
there's still a continuous aspect within
the categorical States
um it's just it's looking at it as a
discrete trajectory I didn't know if
anyone made anything of that found it to
be useful for anything or anything
they're working
oh thanks
um Darius than anyone else
yeah it's not so much a comment on the
previous Point
um about discrete versus continuous
variables I wanted to ask really
a question more
uh more concerned with the mechanisms
behind prediction coding at least as
it's embodied in the brain
um
so you I gotta understand this idea of
sort of the shuttling up of prediction
errors and the shuttling down of
expectations I guess Daniel my question
is
we kind of just take this idea that
prediction errors arise for granted
um something I've been trying to kind of
wrap my head around is
to what degree is that a kind of just um
immediate mismatch between the priors at
one layer and the incoming sense data or
is it that those there's some Bayesian
inference
required to even compute the prediction

error
so you know I have a prior x i compute
variable base to work out the
probability of my prior given y
is that in a sense that distinction
between the prior and the posterior is
that what is constituting surprise or is
it just a pure sense data coming up the
chain which then is discordant with my
prior X if that if any of that makes
sense
yeah
what a lot of uh a lot of angles of this
just first very briefly yeah this uh
hierarchical hybrid type of model where
the higher levels of the system are
discretized
and then the lower levels are kind of
like where the rubber hits the road is
being more continuous that's a motif
that comes up like again and again for
example in Ryan Smith's
um at all's uh work live stream 46
um active inference does not contradict
folk psychology this is a common Motif
and a big reason is that counter
factuals are not really so clear in a
continuous setting
like if there's um three discrete
choices you can really quickly talk
about the tree of all choices
but if it's just like a cone of options
you could take like an infinite amount
of paths because technically even paths
that are only one you know infinitesimal
apart in the continuous space are still
distinct trajectories so in the
continuous time you don't really get a

kind of explicit planning or explicit
counterfactual
and so in order to deal with cognitive
decisions which which often are um
categorical or discrete or deal with
like explicit symbolic actuals then you
need some kind of discretization and yes
the good news is that they work well
together and that gets Revisited in
chapter five
in practice when all of the difference
neural models are going to get brought
together and shown that there's no issue
with the continuous uh model here in the
discrete model up here and then also
comes back in chapter eight
okay
about mechanisms of predictive coding in
the brain well
it's interesting to look at the first
words in this section
chapter 4 gave us a lot of general
information about variational inference
about generative models
um essentially
following from free energy principle in
this chapter we focus on the process
theories
which are are explicitly about a system
of Interest so whereas a principal can't
be falsified or not it's just it's it's
something that is it
um in the realm of decisions that you
take towards modeling
process theories
are are
more in line with scientific hypotheses
they're actually like kind of living and

dying with empirical evidence
but the amount of empirical evidence for
any given statistical model of brain has
nothing to do with um whether the the
principle is valid or not now that
doesn't mean it it obviously is it's
good when we see empirical support for
process theories under the principle
but there's no piece of evidence that
could be found I mean maybe this is an
overstatement that would validate a
principle because principles aren't
validated by evidence but nor are they
rejected by evidence however it's super
interesting to to connect them to
specific systems so
we're going to be applying generative
models
to the brain Bayesian brain plus action
and one of the kinds of models but but
not the only kind of generative model
that exists for the brain is like the
hierarchical predictive coding
architecture and so in that architecture
um which arose from video compression
but then was later found to capture a
lot of biological phenomena
there's an alternating
um stack
of
error units and kind of estimator units
so that the different levels of
estimator units are sending
um ascending error signals and
descending
predictions
so Darius you're you're asking
is that mismatch

immediate
or is it computed through a Bayesian
update
I mean is that the question
yes basically
well the trivial deflationary answer is
Bayesian updates happen in Bayesian
statistics so
if you want to make a statistical model
that calculates an exact mismatch
or if you want to have it do basing
updating
that's about the map not the territory
so we're already lifted away from what
the system's already doing
um in this chapter
in the motor
unit here
it is just
a direct mismatch
descending predictions
of the of the mean
incoming proprioception
and it's just a simple differential
but that simple but
there's no reason why you couldn't make
it slightly different
um like you could have I mean you you
could do anything with Legos
that doesn't mean it would be more
effective or or give unique explanations
or predictions
but in its simplest form
which is embodied here
it is just a differential
denominated in units of the measurement
could you do a Bayesian update
I mean could uh

could this variable
what what would that do or what would we
get by having this do a Bayesian update
I mean I sense the difference between
the two models would be one
you have
predictions coming down the stream
and you get the mismatch coming from
what the higher levels are expecting and
it's kind of meeting at this point where
the prediction error is occurring the
other one is that the prediction
occurring is just happening within a
level where there is a prior
that prior is then getting updated given
evidence and there's going to be a
mismatch between the posterior and the
prior so there's that's all happening
within the level the prior of the level
and the posterior of the level whereas
the other way of thinking about it is
the kind of Prior of a higher level the
proximate level
and that's shuffling down you get
prediction now shutting up but the kind
of level where it's all happening is
just a meeting point under that model
yeah I see what you're saying so like if
if this were instead of just a simple
simple differential If This Were Like a
Coleman filter so it had its own inertia
and it was yeah it was doing it it was
getting input and then it was doing some
like filtering just like common filter
and then it
was sending
some signal back up yeah there's an
infinite amount of model architectures

you could you could have any you know
you could compose it however you want
in principle and no piece of evidence
can support or reject the in-principle
construction or composition
so there are definitely some interesting
properties of that and then this is
where we get to the process theories so
then we look at specific measurements or
or cognitive or behavioral phenomena in
biological systems for example
and then we seek to compose something
that
um captures some of the phenomena that
we want to capture
recognizing that maps and territories uh
are not the same thing
and then you have a whole portfolio of
models and you compare them in terms of
their
um accuracy explainability adequacy
all of these standard statistical
modeling methods
then you write the paper with all the
compared models and you publish the code
then other people continue developing
so
that's kind of more normal science than
it can sometimes seem
but the earlier chapters talked a lot
about what what could be done I mean and
other literature goes goes beyond that
and then they're they're not saying this
is how it is
they're um
it's not the final word on process
theories for active inference it's
simply the interpretation consistent

with available evidence
so they're just saying here's one model
architecture
that we're using for Unique explanations
and predictions
in these different neural systems
now those specific model architectures
are at the mercy of data
like if someone said I think these three
factors contribute to this health
outcome
that's a hypothesis and then someone's
good thing oh no I think it's just these
two and someone else says I think
there's a fourth Factor those are all
going to be compared
just like regular scientific modeling
whereas none of those discussions are
going to be like well the the
three-factor model doesn't fit the
health condition so linear regressions
are incorrect
that's like a category error
but now in chapter five we've kind of
crossed the river from
the General High Road low road
to now I guess I don't know specific
cars on the high road the low road
something like that
um
so this gives us a very different
like flavor and deployment of active
inference
because we're so directly talking about
process theories
and these are the ones that are in
contact with the empirical
any other thoughts or questions on this

okay well chapter five is going to
um go through three
neural systems
from the mammalian nervous system
foreign
and in the end it's going to be kind of
a frontal cortex
a midbrain region
and a spinal cord region
and the chapter is going to introduce
these three different models
which
essentially
um cover some
connected the distinct
cognitive phenomena
and those three systems are going to be
considered separately
and then it all gets built up to figure
5.5
which composes these three systems
and starts getting you towards that
whole iguana
so this is like how how
um functional composition
for these kinds of generative models
works
we're not going to go into the whole
category Theory elements of
compositionality here
but it's just encouraging to know that
the generative models that somebody
makes
um are are are able to be synthesized
and and play nicely together so you
could also compose one of the the brain
and you could compose one of a computer
and then that would be your human

computer interface or you could have two
humans and then that would be your
communication system and
um so that that kind of model
compositionality
is a is a very nice feature that chapter
five makes kind of an understated
argument in support of
hi Ali
okay
5.2 micro circuit semesters any any
overall thoughts or anyone want to like
give a summary on what they read or what
they were left with from this section
um just to comments uh and this kind of
follows from something we briefly
discussed last week in chapter four and
one of the meetings but I found it very
interesting the idea that
um
that they're kind of like asymmetrical
frequencies implied in this 5.2 section
um that that you can have uh like
non-linear functions in the lower level
Computing uh you know errors related to
incoming sensory information versus a
prediction
um and that's synced up in the in a
narrow Imaging that seems to be commonly
associated with like a gamma band
frequency versus uh the descending
predictions that are made which tend to
be at like a slower rate like uh
somewhere around Theta I suppose and um
so it's just it felt like this was
giving a little bit more concreteness to
predictive coding and saying like there
are differences in frequencies there are

various functions or differences here
it's not as simple as um
taking a simple Vector of
the predictions from the upper layer and
the
sensory information like and just doing
some simple subtraction right like the
actual minus the predicted it's rather
that there there's different speeds of
information movement here and different
components different parts and different
ways you could potentially model it
awesome yeah and one short note a lot
all out there and bring in to the notes
um the reactive message passing Paradigm
which I I hope we'll hear more about
from Dimitri and Bert in the Symposium
um these are computational
implementations where different nodes in
the graph can update at different
frequencies so it's perfectly amenable
to that Darius
yes I mean it's um it's in this section
so it's page
87 yes yeah you were just there
um
if you go down a little bit yeah this is
that where it is
um
it's above
the middle schematic oh then yes perfect
so if you just live with that
um I think this is exactly what I was
talking about which is it says here
um in the sort of the middle part of the
bottom paragraph you're sending output
from layer 3 superficial pyramidal
pyramidal cells blah blah blah so the

that layer is shooting prediction there
up then you get the descending input
representative prediction from the
higher level presumably
and while descending output is a
prediction for the lower level
so I think that is kind of maybe worth
unpacking
because
it's I mean
there seems to be a lot going on
um and a lot of predictions going down
and
prediction error going up so it would be
good just to get some clarity on exactly
where all this is happening and
you know what layers we're exactly
talking about here
yeah
yeah
good good point there's a few things so
first I'll bring in one one blog post
that kind of contextualizes like why are
we focusing on this cortical
architecture at all
um Jeff Hawkins
has a a book and and I guess some
startup or something related to this
cortical architecture so
a lot of people have um whether they've
looked at the gross neuroanatomy or
they've looked more into the finer scale
neuroanatomy a lot of people have
centered on
the the cortical regions of the brain
which it's in the mammalian frontal part
of the brain and it's the part that's
wrinkly and on the outside that that

type of tissue organization supports
different cognitive phenomena that
people are interested in of course not
to say to other regions of the brain are
not also doing different cognitive
phenomena but that for certain kinds of
cognition understanding the connections
within and across the stereotypical
units of the cortex called columns this
has been a Super Active area of research
now a really important note
is that while there are six layers
so-called six layers in this in the
cortex these six layers
are histologically defined basically so
they're defined on tissue properties
like what you can see under the
microscope and like gene expression
things like that
this is not a six layer sandwich with
prediction error prediction error six
times so this is not a six level
hierarchical model
this is actually a uh but now this does
have a hierarchical modeling
but if you look at the subscripts or the
superscripts
we see I and I Plus 1.
so actually these six histological
layers together are composing like kind
of
a single multi-scale jump
and then the idea is that across
cortical units because they also signal
laterally not just within the column
um
higher nested models could be
constructed cognitively

but this is not like a six layer nested
model
it just happens to be the case that this
tissue has these six laminar
histological layers
um okay
so the the layers with the numerals
these are these are just
anatomically defined these are not
statistically defined at all
um
and the on the left the anatomy of this
column stereotypical column is shown
and each of these nodes represents like
a cell population it's not necessarily
just a single cell but perhaps like a
group of cells that have some
similarities and like the spdp s s i i
these are like different cell
populations
and this on the left is showing their
empirical
anatomical connections
that's not to say that's all of the
relevant connections just like there's a
subway map and then also there's
footpath so there's bioelectricity
there's neural hormones there's a lot
more in play but this is just the wiring
associated with these cell populations
okay
um
now as common in computational
Neuroscience a given single cell or
population of cells
is going to be associated with some
variable on a statistical model
and here

we basically have um
two
main letters
so first the tilde means Through Time
the two main letters here are the MU
and the um the e
the MU is like the mean estimate
and the e is um the error signal
um
I'm
I'm a little bit not sure why G
is being discussed here
rather than mu
olly do you have a thought on this
uh well uh yeah actually I guess that's
because uh you see based on um songs
framework
um
you see uh all are all the brains needs
or uh the the way that uh the signals
need to be uh passed through both uh
through these uh this bi-directional
path that they don't need to be
satisfied all at once so obviously their
prior prioritized by uh by a
decision-making triangle uh in which uh
the immediate current needs uh or in
other words uh the residual predict
prediction errors
they Converge on uh the as Psalms
describe in his book uh periquidactyl
gray and they're ranked in relation to
uh the uh the current needs and
opportunities uh in and and this
opportunity is manifested in a in the
form of a kind of two-dimensional
saliency map uh in the superior
colliculi so

uh this um prioritization uh triggers a
kind of conditioned action programs uh
which uh in turn unfolds in expected uh
context over a kind of deep hierarchy of
predictions or in other words the
general model of the
um of the forebrain
so these actions uh they're generated by
a prioritized uh affects that are
generalized that are generated by
prioritized ethics they're somehow
voluntary uh in other words they're
subject to the immediate Here and Now
choices
right uh kind of pre-established
programs or pre-established algorithms
and on the other hand these choices are
kind of felt in uh if we want to talk
about the higher order
um the higher order elements of the
brain they're kind of felt in the
extraceptive Consciousness which again
contextualizes this kind of affect and
uh they're they're made on the basis of
this fluctuating uh Precision waiting uh
in other words uh arousal modulation
um and postsynaptic gain of the incoming
error signals uh that are on the other
hand rendered silent by these
prioritized needs and they're uh they're
also buffered in working memory uh with
the aim of minimizing uncertainty or
maximizing confidence uh which also
correlates with the current prediction
as to how that particular need
or particular I don't know desire to
minimize the prediction error needs to
be met so

briefly what this amounts to uh is a
kind of a bit unsettling
which kind of says that
the higher order elements of the of the
brain or the Consciousness would only be
required to mediate mismatches between
the internal and external States and the
more adaptively and predictably reliable
our model becomes the more our Behavior
will become automated uh in other words
operating below the level of conscious
experience
so
yeah I guess that balance between the uh
I mean the Precision making of the gnd E
comes from uh or at least can be
explained from uh psalms's framework in
this way
oh awesome
but there's no G in this figure
well here's G here's the full descending
G
so just be just just like um you know
the knee bones connected to the leg bone
and all of that ascending and descending
are relative to obviously which node
we're talking about
so the ultimate descending
um prediction
again in a in a a standing person the
ultimate descending prediction is going
to be like the the main big output of
this model
and that's what's going to
um
uh it's not shown here but it's it's
what is the the full output of this
whole column is this G

um
I mean there there's so many oh yeah
I'll go for it
no I just wanted to add another uh Point
here uh which uh you see these kinds of
Errors uh well the word error can
somehow
um connote negative meet uh I believe in
the free energy principle framework or
active inference framework or at least
Mark zomes's interpretation of it uh
that's exactly what our Consciousness
feeds on so
um in other words
uh
the surprisal uh I mean uh the
Consciousness as I said is only needed
uh I mean roughly speaking in situations
where surprisal needs to be minimized uh
if
no surprise though there would be not
much need for Consciousness or to put it
in a different terms at Consciousness is
a kind of process that
seeks to somehow overcome itself so
these kinds of error are essential for
uh for any higher order activity of the
brain be it I mean Consciousness
decision making or any kind of higher
order activity so it would never be
exactly uh equated to zero because that
would basically mean uh to go
unconscious and to not have any higher
order activity uh in the brain so this
also
maps onto annual Seth's Theory Of
Consciousness as to uh as a kind of
process that um seeks to minimize the

surprisal
yeah also error
um yeah like you said Eric can seem like
it's a bad thing but it's like a set
point and half the time
your error is going to be well
in a calibrated system half the time the
error is positive and half the time the
error is negative if you're spending
more time
if ninety percent of the time
your estimates are too high
in a galaxy in the world then that would
be a signal to update
um
there's a lot more detail to go into
with the exact
um
mechanics here
um including
the the difference between the Middle
version and then the version on the
right which is more connected with the
notation that we saw from the pomdp
um but this is kind of like a um
it's like a
for some reason didn't give a fancy iced
coffee percolator
but something is coming in and this is
like a percolation mechanism
and there's a there's a descending
from I plus one
that's the G here
prediction from the higher level
and then there's the um ascending
uh differential this isn't the raw
signal data coming in this is already
the differential coming up

and then those are able to meet
through a statistical
mechanism
that also recapitulates but but not
exactly I mean it's not like it's this
five and three it's it's it's not
exactly intended to to be
but this is the result of the this is a
falsifiable process Theory
that then somebody can explore like well
what statistically happens when this
when you transiently suppress this exact
connection
and then the process of model building
continues Darius
so I asked whether the I in this
um is that referring to the tildew is it
referring to time where I plus one is uh
future or past predictions and whatever
or is it actual is it part that is it
relating to the level of the hierarchy
correct I is the ith level of the
hierarchy I plus one is a higher level
the hierarchy tilde means Through Time
um
now they're going to abstract
a little bit from this six layer showing
all the nodes
so here we still have one two three four
five six with figure five or I mean uh
with a uh layer four
um being shown
now it's being clarified how
things falling in my room
now it's um being clarified how here we
have I minus 1 I and I plus one
so these are three cortical columns that
are next to each other

and now those three cortical columns are
able to actually instantiate
a three layer nested model
so it's almost like we get the vertical
stacking through horizontal uh
concatenation or composition
where the outputs of one column can be
errors
like here from from I
um and also notice that I that they're
not bound within a column
here's I minus one
and then the error unit of I
then the mean
and the error units of I
and then the error units of I plus one
and then the highest nested layer on the
right
[Music]
um
which is uh only sending descending
predictions and getting the highest
level ascending predictions
yeah I'm just a bit closed with what you
just said uh
in terms of you know uh what I I minus
one and I plus one mean here so formula
works you know three different time
steps uh within the the tilde uh you
know big Vector of of states
uh and not you know a communication
between different columns
yeah the the I the superscript is not
time
subscript is used for time
but here this is referring to three
nested levels so let's just say this
cortical column we're guessing the

hundreds place this one where the tens
place and this were the ones place
so together this is a three layer nested
model
that's
um doing predictive processing on the
three-digit number
but time is not shown here it's just
saying tilde it's all happening through
time
and here the I superscripts are related
to the level of the hierarchical model
okay
Darius
sorry I don't know why it keeps going
into clapping
um
I do mean to clap everyone's making
excellent contributions um
um it's so just to summarize it seems
like there's hierarchy both Within
and across
so if you go back up to that
it so the way you spoke of that before
is this is a free layered nested
hierarchy but we also have superficial
and deep down the vertical
so
what I mean it's a very general question
but why is this computationally
efficient to have both
vertical and horizontal hierarchies why
can't we just shuttle predictions and
prediction error across one dimension
yeah well I I stated this though it has
six tissue layers
is really like is this is like a minimal
multi-scale model

um you could make a model
you could make a predictive processing
hierarchy that's like a skyscraper and
just simply has three
on top
but it just turns out that this is
actually capturing the structure of the
brain
so I guess what I mean is what we what
do we what's what's gained
by I I you know it makes complete sense
I mean whether it's vertical horizontal
I think it's a moot point because it's
just it's just way of I guess picture I
guess but what's gained by having
two Dimensions rather than one so what I
mean by that is what what is the
function of the verticality the deep and
superficial verticality
if ascending and descending product
predictions is compensated for or
counted for by the horizontal axis in
this picture in this diagram
yeah one answer is that the the the
histological
um verticality let's just say the fact
that there's multiple layers of tissue
gives you a really nice packet or motif
that has
um both the units that you need the MU
and the E so it has the two pieces that
you need so you could have your workshop
with all the with just a bunch of mu's
and E's
and then just compose it however you
needed it um
without caring if you're reusing motifs
but if we construct a motif like a

cortical column
that that's able to
um do what it does by itself and also
um contribute to more nested models
just through a diagram
and combining them like this
it just makes a scalable reusable motif
um that's developmentally plausible and
actually has support in neurobiology but
then Javier
yeah
isn't the the horizontality here mustn't
think about different areas in the brain
that could also be seen as a hierarchy
you know vertical hierarchy between you
know a lower area like you know V1 V2
and V4 in the visual system
so we will have you know both
verticality within uh you know each of
this area and you know a more General
connection between areas in which you
you know you have a more and more
complex uh environment message
yes
you could coarse grain and you could um
look at other neural systems
in terms of message passing and
hierarchical models
go for it heavier yeah
um connecting back with a thousand uh
brain's theory of Hawkins wouldn't
disconnect as well with the fact that
you know Hawkins says that you have you
don't have this one model you have all
these columns and micro columns like
creating lots of different models and
then you have this sort of voting system
that is connected is related to the

horizontal Connections in which the
different columns are influencing each
other is that a sort of way of
explaining this you know vertical versus
horizontal so maybe the horizontal sort
of voting or influencing uh the models
created by the different
columns or micro columns
yeah that's a great
a great um comment one paper that kind
of approaches that is this Fame in the
brain Fame in the predictive brain
um Dennett had this Fame in the brain
meme
um
and and uh
this paper argues that basically like
the the unit the unity of consciousness
is like kind of like underlying it
there's this subconscious process of
selection amongst those models
I mean again this is this this is a is
running fast and loose with the map in
the territory because we're talking
about a system of interest but we're
obviously talking about the acting
funtology but like if um 10
counterfactuals are imagined
each one of them in deliberation might
be transiently famous and and explicitly
considered like you know I could go to
coffee shop a I could go to B I could go
to Z
but then for things that are experienced
that can be understood as like the
outcome of this election or selection
process
hmm

so it's like the winning the winning
hypothesis
is what is you know in the spotlight of
Consciousness and or awareness at a
given time Ali
yeah and also aside from that saliency
map uh that uh I mentioned before I
don't believe uh the representation of
the nested hierarchy uh is uh is about I
mean a necessarily
um uh it's not necessarily spatially
distributed or I mean horizontally
distributed uh representation it's more
like uh the unpacking of a single uh
let's say Tangled uh neurons that's
unpacking that figure so the real
physical or
um spatially distributed hierarchy is
the one we we have uh uh the the
vertical one in which the clusters of
different types of neurons
are packaged into uh I mean vertical
columns so and and that's also I believe
uh so Darius asked what's the uh
computational justification for
um such an organization well that kind
of packaging can be efficient if we
consider uh the different types of
neuron with a different length of axons
for each neuron so
uh so each of those types of neurons can
be assigned with
specific tasks or can be more efficient
in doing some tasks other than the
others so uh yeah I believe that's also
one of the reasons for the vertical
distribution of those types of neurons
but again that nested hierarchy is just

schematic representation or unpacking of
a single
Tangled kind of neurons and not
necessarily it doesn't
refer to any spatially distributed
hierarchy
yeah also we can think back to the neats
and the scruffies I mean the extreme
scruffy position is every synapse needs
its own framework and University
department and software package
on the other end you have the extreme
neat framework and and so real work
exists on that Continuum and that's part
of the unfolding discussion
and one can only imagine how much
neuroimaging and and molecular histology
goes into the development of these
really elegant computational
um models and then
people take them in different ways some
people take these in abstract or
generalize them even more and aren't as
worried about whether it actually
adheres to the structure of the brain
um like you know what if we had two
cortices on top of each other and we had
this and we had that
um someone else might want to
um make it more directly connected to
the measurements they were taking in
their lab
so again chapter five is is a it's a a
five switch from chapter four because
chapter five
really is getting into the empirical and
the specific systems of Interest that's
going to come back again in chapter six

when we talk about the recipe and um
chapter nine when we talk about
integration with data
um
but that's kind of the the breadth that
that we're in
between on one hand the more theoretical
work like free energy principle and
active inference in principle and on the
other hand these are specific models and
and
[Music]
um
what do you call somebody who makes an
evidence-based critique of this model a
computational neuroscience
student or researcher I mean that that's
all there is to that field
looking at how other people have um
integrated the information and made
models and then offering
um positive developments and critiques
beyond that
so it's kind of interesting after all
like the math and and philosophy that it
can feel like in the earlier chapters
here which is like dumped squarely into
empirical neuroscience
and the connections um
just signaled by a new chapter
just so that we see it before we like
write down our questions and and
hopefully
um you know people can add as much
discourse it's super super helpful to
vote or add
um your own thoughts and curate the
answers of others

um in the coming week but basically
here's what's going to be upcoming
we're going to go from looking at the
cortical column
to descending predictions
playing a role in motor control
proprioception proprioceptive
information is going to come in and then
in the ventral and dorsal horns but
basically in this cross section of the
mammalian spinal cord the descending
predictions are going to meet with the
proprioceptive incoming information and
that's going to result in kind of this
differential guided
motor activity selection like
I want my arm to be here
it is where I want to be it's not moving
move the set point now the r moves
then
um
going back up to the cortical column
we're going to now Trace another
projection so not just a projection to
the spinal cord but also cortical layer
5 targeting some
deep brain structures like the stratum
and other aspects basal ganglia
very interesting very cool dopaminergic
and and brain regions
uh
several
topics and cognitive phenomena are
discussed
that the midbrain regions are important
for
here in figure 5.4 we also talk about
policy selection and the direct and the

indirect Pathways of policy selection
one of which the indirect is basically a
pass-through on habit
and the other the direct pathway
involving an expected free energy
calculation so thus updating prior
probabilities on policies
habit updating that
according to the expected free energy of
policies which is to say their epistemic
and pragmatic value so this is kind of a
type 1 type 2 or like a habit
pass-through versus deliberative
decision making distinction
table 5.1
provides some empirical work connecting
different computational
variables to neurotransmitters
of course neurotransmitters do different
things in different brain regions
different synapses different contexts
this is the neat and Scruffy again we
could have a framework for every single
specific neurotransmitter or we could
just say all neurotransmitters are the
same and this is somewhere in the middle
this is saying we can actually
understand some some uh uniquely
valuable insights and unique predictions
from associating different
neurotransmitters in certain contexts
with different
computational roles
5.6
comes to this hybrid
theme
which is about connecting the continuous
and the discrete hierarchies it's a

total major theme of the textbook
though we are often distinguishing
between discrete and continuous State
spaces
and also a special focus on the
continuous and discrete treatment of
time
as with another modeling
um
it's it's Revisited multiple times
that you can make Hybrid models for
example with continuous modeling at a
lower level and category Oracle modeling
at a higher level
um finally in the closing pages
they stitch together
all three of the systems that were
introduced
so the descending signaling
coming from
uh the cortical column
on one hand going down to the motor
selection unit
however
also going down to the behavioral
planning and selection components and in
fact those behavioral selection and
planning components
then re-entering
into
[Music]
um
the cortical
so that the action can be decided
so just one example of uh
compositional cognitive cartography
and bringing together
different modeled subsystems of Interest

and using active inference to integrate
diverse cognitive phenomena
and that's chapter five
any last comments
well
next week will be our uh
it's the Symposium so hopefully people
can view it or or re-watch it
then uh it'll be our final discussion on
chapter five
and the following week will be an
unrecorded session for for like feedback
um we really need everyone
truly so we will be especially
interested to hear everyone's feedback
with this Anonymous feedback form
for people's um suggestions on how to
develop and improve there's many many
notes
so it it for those who want to stay
active and and leverage their
contribution
reading what people have already
suggested which is quite copious
and then curating or adding it and then
maybe even finding a place where like
you want to continue to get more
involved in the coming months and years
then we'll uh take some weeks off
and then restart the second half of
cohort four with chapter six
meanwhile start cohort five from the
beginning of the book
um
and uh
the work goes on
okay
thank you everyone see you next time

thank you bye thank you